

3.6 Intermolecular forces homework — answers

CIE 9701 AS Chemistry · 8 multiple-choice + 10 written marks

Do not issue this file to students.

Section A — Multiple choice

Quick answers

Q	Ans.	Q	Ans.	Q	Ans.
A1	C	A4	B	A7	B
A2	C	A5	B	A8	D
A3	B	A6	A		

Worked explanations

A1. C

2021 FM Paper 12 Q5

Temporary dipole–induced dipole forces (instantaneous dipole–induced dipole) are the *only* intermolecular force in a substance whose molecules have *no permanent dipole* and *no H-bonding*. Octane, C_8H_{18} , is a non-polar hydrocarbon, so this is true for octane.

A: hydrogen chloride is a polar molecule, so it also has permanent dipole–permanent dipole attractions.

B: methanol, CH_3OH , has an O–H group, so it shows hydrogen bonding.

D: water has extensive hydrogen bonding.

A2. C

2021 MJ Paper 13 Q5

Substance 1 has permanent dipole–dipole attractions → a polar molecule with no H-bonding: CH_2Cl_2 (polar C–Cl bonds; no O–H or N–H). Substance 2 has only weak forces between molecules → a non-polar molecule: CH_4 . Substance 3 has hydrogen bonding → a molecule with an O–H group: CH_3OH . So substance 1 = CH_2Cl_2 , substance 2 = CH_4 , substance 3 = CH_3OH .

A / B / D: mis-match at least one substance — e.g. Al_2Cl_6 is non-polar (it would give only weak forces, not permanent dipole–dipole), and H_2O in the “substance 3” column would leave CH_3OH unplaced.

A3. B

2024 MJ Paper 11 Q8

Down Group 16 the hydrides H_2S , H_2Se , H_2Te show a steady increase in boiling point because the molecules get larger with more electrons, so the van der Waals’ forces increase. Water breaks the trend because the O–H bond allows *strong hydrogen bonding* between water molecules, which is much stronger than the van der Waals’ forces in the other hydrides.

A: shielding/nuclear charge does not explain a *higher* boiling point anomaly of this size.

C: the covalent O–H bonds are not broken during boiling; only intermolecular forces are overcome.

D: smaller molecules have *weaker*, not stronger, van der Waals’ forces.

A4. B

2024 MJ Paper 12 Q8

Identify the strongest intermolecular force in each compound: 1 $(\text{CH}_3)_3\text{CH}$ (2-methylpropane) — non-polar, only van der Waals' forces (lowest bp, -12°C). 4 $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ (butane) — also non-polar, slightly larger so slightly stronger van der Waals' forces (-0.5°C). 3 $\text{CH}_3\text{CH}_2\text{CH}_2\text{SH}$ (propane-1-thiol) — weak permanent dipole–dipole (67°C). 2 $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (propan-1-ol) — hydrogen bonding (97°C). Increasing boiling point: $1 < 4 < 3 < 2$.

A / C / D: each reverses part of the order — e.g. placing the alcohol below the thiol, or the thiol below butane.

A5. B

2021 ON Paper 12 Q6

Compare the overall dipole moments: CO_2 is linear and symmetrical, so its bond dipoles cancel and the overall dipole is *zero*. The other three all have a permanent dipole, and $\text{H}_2\text{C}=\text{O}$ (methanal) has the largest — the very polar $\text{C}=\text{O}$ group with no other polar bond to oppose it ($\mu \approx 2.3 \text{ D}$, larger than $\text{CH}_2\text{Cl}_2 \approx 1.6 \text{ D}$ and $\text{COCl}_2 \approx 0.9 \text{ D}$).

A: CH_2Cl_2 is polar, but its dipole is smaller than that of $\text{H}_2\text{C}=\text{O}$.

C: COCl_2 is polar, but the two $\text{C}-\text{Cl}$ dipoles partly oppose the $\text{C}=\text{O}$ dipole, so its overall dipole is the smaller of the polar molecules here.

D: CO_2 is symmetrical, so it has *no* overall dipole.

A6. A

2021 ON Paper 11 Q33

All three statements are correct. (1) Ethylene glycol molecules insert into and disrupt the extensive *hydrogen-bond network* of ice, so it melts below 0°C . (2) Ethylene glycol has two $\text{O}-\text{H}$ groups, so its molecules hydrogen-bond to each other. (3) It also hydrogen-bonds to water molecules, so it dissolves in the meltwater. Statements 1, 2 and 3 are all correct.

B / C / D: each rejects at least one correct statement. Statement 1 is the key one: de-icing works precisely because the glycol disrupts the H-bond network of ice.

A7. B

2021 MJ Paper 12 Q3

Each compound has four carbon atoms: X = butan-1-ol ($\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$) — hydrogen bonding. Y = ethoxyethane ($\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$) — permanent dipole–dipole (an ether, no $\text{O}-\text{H}$, so no H-bonding). Z = sodium butanoate ($\text{CH}_3\text{CH}_2\text{CH}_2\text{COONa}$) — ionic bonding. The stronger the force broken on boiling, the higher the boiling point: Y (pd–pd) $<$ X (H-bond) $<$ Z (ionic). So lowest = Y, highest = Z.

A / C / D: these rows put the compounds in the wrong order — e.g. placing the ionic compound Z below the alcohol, or the ether above the alcohol.

A8. D

2025 MJ Paper 12 Q10

(1) **False** — both ΔH_b (vaporisation) and ΔH_m (melting) are *endothermic*, so they are *positive*, not negative. (2) **True** — vaporisation breaks all the intermolecular forces, while melting only loosens the lattice, so $\Delta H_b > \Delta H_m$. (3) **True** — ice and water are both held together by the same type of force, hydrogen bonding. Correct statements: 2 and 3 $\rightarrow$ D.

A / B / C: all include statement 1, which is false because the enthalpy changes of melting and boiling are positive.

Section B — Mark scheme clips

B1 [1]

1(b)(iii)	hydrogen bond	1
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B2 [2]

1(b)(iv)	M1 link shown as a dashed line between the lone pair of electrons from N of one NH ₃ to one H on other NH ₃ M2 minimum 3 correct partial charges (on adjacent atoms) over two NH ₃ molecules EITHER $\delta^- \text{N} - \delta^+ \text{H} \cdots \delta^- \text{N}$ OR $\delta^+ \text{H} \cdots \delta^- \text{N} - \delta^+ \text{H}$	2
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B3 [2]

2(c)(iii)	N ₂ (g)	CO(g)	2
	instantaneous dipole–induced dipole ✓	instantaneous dipole–induced dipole (and) permanent dipole–permanent dipole ✓	

B4 [3]

1(d)(i)	SO ₂	–73	pd	3
	H ₂ O	0	H	
	SiO ₂	1600	C	
	Na ₂ O	1132	I	
	MgO	2852	I	

B5 [2]

1(a)(ii)	M1 bar at HF shows any boiling point above HI on graph M2 explanation of difference in boiling point of a sample of HF in terms of strength (sum of) hydrogen bonds (and induced dipoles in HF) are stronger than (sum of) induced dipoles (and permanent dipoles in HCl / HBr / HI)	2
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